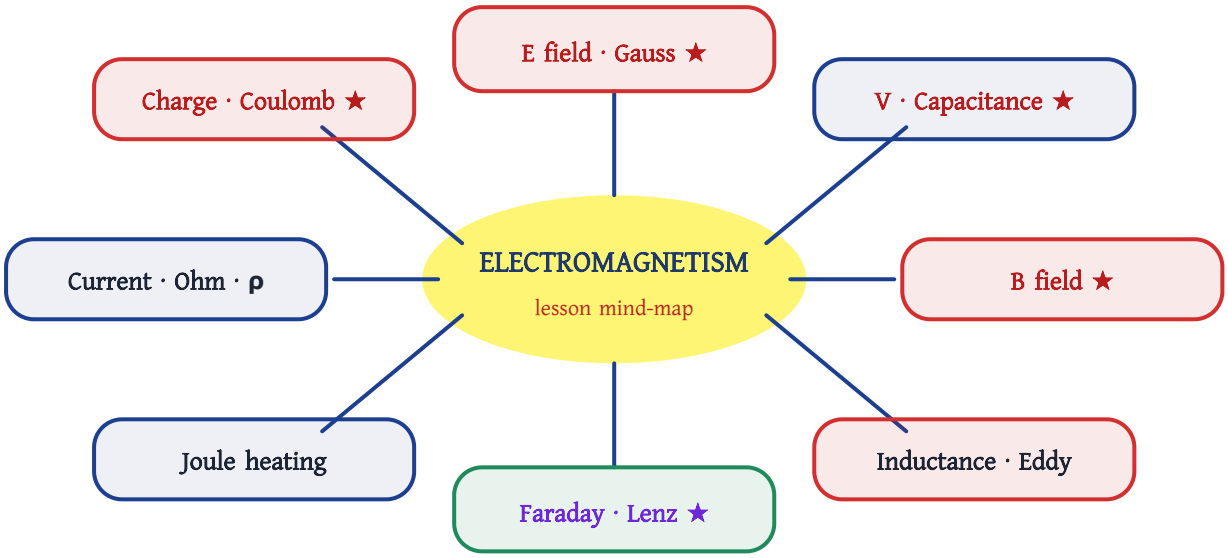
Electromagnetism

Charge · Gauss · Current · B · Faraday · Maxwell: exam-ready in 11 sheets



↑ the whole lesson on one page: red = high-yield, green = applied / extra

Syllabus coverage

"Basic elements: **charge**, **Coulomb's law**, electric field, **Gauss's law**, electric potential, magnetic field; **Faraday's law**, **Maxwell's equations**". This notebook covers that whole line at university depth: charge · electrostatics (field, Gauss, potential, capacitance) · steady current (Ohm, resistivity, Joule heating) · magnetostatics (Biot–Savart, Ampère, Lorentz force) · induction (Faraday, Lenz, self & mutual inductance, eddy currents) · Maxwell's equations with EM waves. Neighbouring lessons: *Waves and Oscillations*, *Special Relativity*, *Quantum Physics & EM Waves*.

High-yield list

① k, ε_0, μ_0, e, c
② Gauss $\Phi = q/\varepsilon_0$
③ $E \propto 1/r^2$, $V \propto 1/r$
④ $R = \rho L/A$ · $H = I^2Rt$
⑤ $\varepsilon = -N\, d\Phi/dt$, Lenz
⑥ Maxwell's 4 laws

Legend: **yellow** = key term · **red circle** = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

Contents

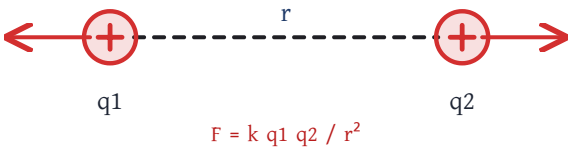
✓ Charge, quantisation, Coulomb ..... p. 2
✓ Electric field & Gauss's law ..... p. 3
✓ Potential & capacitance ..... p. 4
✓ Current, resistance, resistivity ..... p. 5
✓ Magnetic field, Biot–Savart ..... p. 6
✓ Flux, Faraday, Lenz, emf ..... p. 7
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1 Electric Charge & Coulomb's Law

Definition: **Charge (q)** is the property of matter that makes it experience an **electric force**. It is a scalar; only two kinds exist, **positive (+) and negative (−)**, and the SI unit is the **coulomb (C)** = 1 A·s.

- **Quantised:** $q = \pm ne$, $e = 1.602 \times 10^{-19}$ C — no fractional charge
- **Conserved:** total charge of an isolated system never changes; rubbing only **transfers** it
- **Invariant:** the same in every inertial frame (unlike relativistic mass)

> sign first, then magnitude!



like charges push apart along the joining line

Coulomb's law (1785): $F = k\,q_1q_2 / r^2$

$k = 1/4\pi\varepsilon_0 = 8.99 \times 10^9$ N·m²/C²

$\varepsilon_0 = 8.854 \times 10^{-12}$ C²/(N·m²)

F in newton, r in metre · attractive if unlike

You change...	Force becomes	Reason
r → 2r	F/4	inverse-square law
r → r/2	4F	same law, read backwards
both charges ×2	4F	$F \propto q_1q_2$
one charge ×2	2F	$F \propto q$, not r

In a medium $F = F_{vacuum}/\varepsilon_r$; water ($\varepsilon_r \approx 80$) cuts the force about **80 times**.

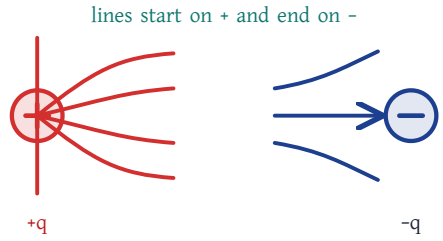
Memory trick: Coulomb is **central** and **inverse-square**. Memorise $k = 9 \times 10^9$, not ε_0: every exam numeric uses 9×10^9.

Worked example 1 ✍️: +2 μC and +3 μC, 0.3 m apart, vacuum:
$F = 9 \times 10^9 \times 6 \times 10^{-12} / 0.09 =$ **0.6 N, repulsive** (both positive).

Exam trap: $k = 9 \times 10^9$ is **$1/4\pi\varepsilon_0$**, not ε_0. Write $F = 9 \times 10^9\ q_1q_2\ \varepsilon_0/r^2$ and the answer is out by $\sim 10^{10}$.

2 Electric Field (E)

Definition: The **force per unit positive test charge**: **$E = F / q_0$**. E is a **vector**, SI unit **N/C = V/m**; for a point charge **$E = kq/r^2$**.

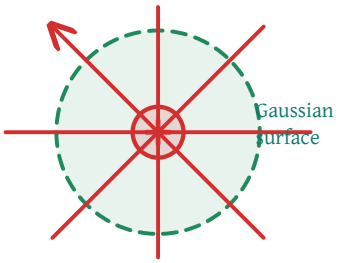


lines never cross and never close

- Lines leave **+**, land on **−**; density ∝ E
- They **never cross** — E has one direction per point
- They meet a conductor at **90°**; **E = 0** inside it
- Analogy: E = F/q ↔ g = F/m

3 Gauss's Law

Statement (1835): The total electric flux through **any** closed surface = enclosed charge ÷ ε_0: $\oint \mathbf{E}\cdot d\mathbf{A} = q_{enc} / \varepsilon_0$ (flux $\Phi_E = EA\cos\theta$, unit $N\cdot m^2/C$).



- Flux depends **only on the charge inside** — not on the size, shape or position of the surface
- Charge outside → net flux **0**: as many lines enter as leave
- E leaves the integral easily only for **symmetric** distributions, where E is constant and ⊥ dA

Charge distribution	Field E	Remember
Point charge / outside a sphere	kq/r^2	acts as if all q were at the centre
Infinite line charge (λ C/m)	$\lambda / 2\pi\varepsilon_0 r$	∝ 1/r, falls off slowly
Infinite plane sheet (σ C/m^2)	$\sigma / 2\varepsilon_0$	same at every distance
Just outside a conductor	σ / ε_0	twice the sheet value
Inside a conductor	0	charges migrate to the surface

Memory trick: **Sphere → $1/r^2$, line → 1/r, sheet → constant.** Count the dimensions you can spread a charge over: 2, 1, 0.

Classic MCQ: A charge q at the centre of a cube → flux through **one face** = $q/6\varepsilon_0$: the total q/ε_0 is shared equally by 6 faces.

Worked example ✍️: E at 20 cm from a 2 μC charge: $E = 9 \times 10^9 \times 2 \times 10^{-6} / 0.04 = \mathbf{4.5 \times 10^5\ N/C}$, directed away from the charge (q > 0).

4 Electric Potential (V)

Definition: Work done per unit positive charge in bringing a test charge from infinity: **V = W/q**. V is a **scalar**, unit **volt** = J/C; for a point charge **V = kq/r**.

	Field E	Potential V
nature	vector	scalar
point charge q	kq/r^2	kq/r
unit	N/C = V/m	volt = J/C
relation	$E = -dV/dr$	

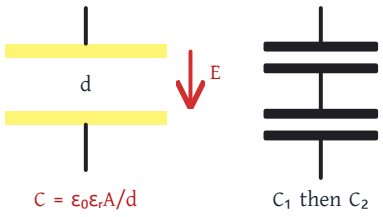
- E points from **high to low** V
- $\Delta U = q\Delta V$; **1 eV = 1.602×10^{-19} J**
- **Equipotentials ⊥ field lines**; no work along one

Worked example ✍️: V at 30 cm from a +2 μC charge: $V = 9 \times 10^9 \times 2 \times 10^{-6} / 0.3 = \mathbf{6 \times 10^4\ V}$. Doubling r halves V, while the **field** would fall to a quarter.

5 Capacitance & Capacitors

Definition: $C = Q/V$: charge stored per unit potential difference, unit **farad (F)** with $1\ F = 1\ C/V$. Parallel plates: $C = \varepsilon_0\varepsilon_r A/d$.

Two capacitors C_1, C_2	Result
In series	$C = C_1C_2/(C_1+C_2)$
In parallel	$C = C_1 + C_2$
Energy stored	$\frac{1}{2}CV^2 = \frac{1}{2}Q^2/C = \frac{1}{2}QV$
Charging to V	$q = CV(1 - e^{-t/RC})$ · $\tau = RC$



V = Ed between the plates

Memory trick: Parallel adds like sums of money; series is the **tightest link** — limited by the smallest capacitance, exactly like resistors.

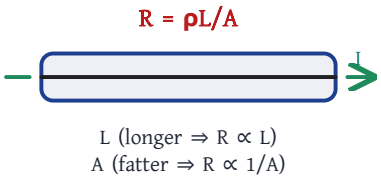
Exam trap: Farad = capacitance, weber = flux, henry = inductance. Two 3 μF in series give **1.5 μF**, not 6 μF.

Worked example ✍️: Air plates, A = 0.01 m^2, d = 1 mm: C = **88.5 pF**; at 200 V, $Q = CV = \mathbf{1.77 \times 10^{-8}\ C}$, $U = \frac{1}{2}CV^2 = 0.035$ J.

6 Current, Resistance & Ohm's Law

Definitions: **Current $I = Q/t$** (ampere, 1 A = 1 C/s) · **Ohm's law $V = IR$** · **Resistance $R = \rho L/A$**, where ρ is the **resistivity** of the material and R a property of that *piece* of wire.

- **L ×2** → R doubles; **A ×2** → R halves; conductivity $\sigma = 1/\rho$
- **Temperature:** $R_t = R_0(1 + \alpha t)$, α quoted at 20 °C
- Joule heat $H = \mathbf{I^2Rt = V^2t/R}$; power $P = \mathbf{VI = I^2R = V^2/R}$

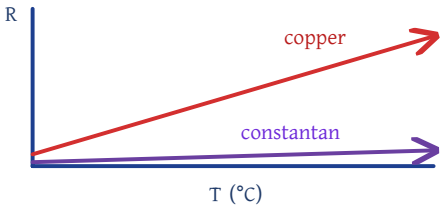


L (longer ⇒ R ∝ L)
A (fatter ⇒ R ∝ 1/A)

a wire is a resistor

Material	ρ (Ω·m), 20 °C	α ($°C^{-1}$)
Silver (best conductor)	1.59×10^{-8}	3.8×10^{-3}
Copper	1.68×10^{-8}	3.93×10^{-3}
Aluminium	2.82×10^{-8}	3.9×10^{-3}
Constantan (≈ 0 alloy)	$\mathbf{4.9 \times 10^{-7}}$	**≈ 0**
Nichrome (heater wire)	1.0×10^{-6}	4×10^{-4}

Worked example ✍️: Copper wire, L = 100 m, A = 2 mm^2: $R = \rho L/A = (1.68 \times 10^{-8} \times 100)/(2 \times 10^{-6}) =$ **0.84 Ω**. At 5 A, $P = I^2R = 25 \times 0.84 =$ **21 W** of heat. Half L and double A → R unchanged.



R rises with T for metals; flat for alloys

Memory trick: **ρ = RA/L** is the material fingerprint: copper 1.68×10^{-8} and silver 1.59×10^{-8} Ω·m settle every "better conductor" MCQ.

Exam trap: R is not a property of the material alone. Constantan and nichrome are chosen for heaters because **α ≈ 0**, so R stays steady as it heats.

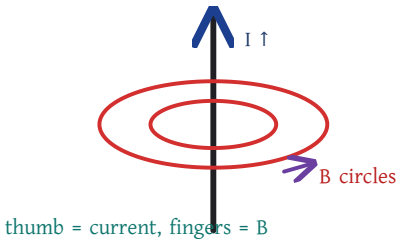
IT link: **V = IR** for a fixed resistance is a straight line through the origin — which is exactly why a two-resistor voltage divider scales a 5 V logic level, and how every SPICE resistor model works. Ohm's law is the only ingredient in Kirchhoff's laws.

7 Magnetic Field (B) & Its Sources

Definition: The region in which a **moving charge or a current** feels a force. **B** is a vector, SI unit **tesla (T)** = N/(A·m) = Wb/m^2, and **1 T = 10^4 gauss** (1 G = 10^{-4} T). $\mu_0 = \mathbf{4\pi \times 10^{-7}}$ **T·m/A**.

Biot–Savart law for a small current element:
$dB = (\mu_0/4\pi) \cdot I\, dl \sin\theta / r^2$

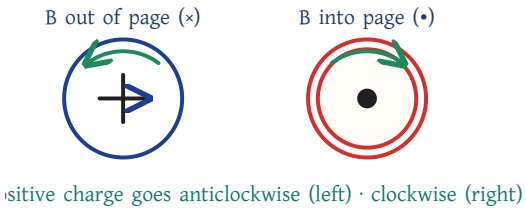
Source	B	Falls off as
Long straight wire	$\mu_0 I / 2\pi r$	1/r
Centre of a circular loop	$\mu_0 I / 2R$	1/R (not $1/R^2$)
Solenoid, inside (n turns/m)	**$\mu_0 nI$**	uniform, ∝ n
Long solenoid, far outside	$\mu_0 NI / 2L$	like one wire, 1/r



right-hand grip rule

- **Straight wire:** thumb = conventional current, curled fingers = B
- **Solenoid:** fingers follow the winding, **thumb points north**
- Like currents in parallel wires **attract**, opposite currents repel

Forces on charges and wires: Lorentz force **F = q(v × B)**, **|F| = qvB sin θ**
Wire **F = BIL sin θ** · coil torque τ = **NIAB sin θ**
Circular path r = mv/qB · ω_c = qB/m



sitive charge goes anticlockwise (left) · clockwise (right)

Memory trick: Fleming's **left** hand = **mo**tor: **F**ield, se**C**ond, thu**M**b. The right hand is the **generator** rule.

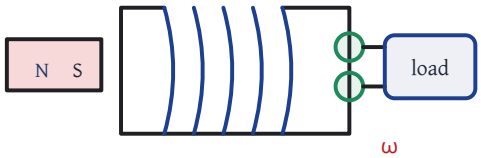
Exam trap: F = **0** when θ = 0° (v ∥ B) or the charge rests, and the magnetic force does **no work** — it bends the path, never the speed.

Worked example 1 ✍️: B at 5 cm from a long straight wire carrying 2 A: B = 2 × 10^{-7} × 2 / 0.05 = **8 × 10^{-6} T**, anticlockwise for upward current.

Worked example 2 ✍️: Solenoid n = 500 turns/m, I = 2 A: B = 4π × 10^{-7} × 500 × 2 = **1.26 × 10^{-3} T** inside. An electron at 10^6 m/s feels F = qvB = **2.0 × 10^{-16} N**.

8 Magnetic Flux & Faraday's Law

Magnetic flux: **Φ_B = BA cos θ**, unit **weber (Wb)** = T·m^2 = V·s. θ is the angle between **B and the area normal** (0° gives maximum flux, 90° none). Faraday (**1831**): a **changing magnetic flux** induces an emf, **ε = -N $d\Phi_B/dt$** for N turns.



slip rings feed the load → AC out

rotating coil: ε = NBAω sin ωt

Lenz's law is the minus sign: the induced current always **opposes the change** in flux that produced it, so no energy is created — this is conservation of energy.

An emf appears when **B**, the **area** or **θ** changes, and also when the circuit moves. A steady flux, even a huge one, induces nothing.

Situation	Induced emf	Note
N-turn coil, flux change	**ε = -N ΔΦ/Δt**	N multiplies the emf
Rod of length L moving at v ⊥ B	**ε = BLv**	motional; no magnet needed
Coil rotating at ω in B	ε = NBAω sin ωt	alternating, ε_{max} = NBAω
Self-inductance	ε = -L dI/dt	opposes a change of *current*
Mutual inductance	ε_2 = -M dI_1/dt	basis of the transformer

Memory trick: Faraday in words: **more turns, faster change, bigger emf.** Lenz in words: **the induced effect always brakes the cause.**

Exam trap: Lenz opposes the **change**, not the field. Push a magnet in and the coil repels; **pull it out** and it still attracts. An emf is volts, **not** newtons.

Worked example ✍️: The flux through a 100-turn coil falls from 0.05 Wb to 0.01 Wb in 0.2 s: $|\varepsilon| = N\,\Delta\Phi/\Delta t = 100 \times (0.05 - 0.01)/0.2 = 100 \times 0.04/0.2 =$ **20 V**, in the direction that opposes the flux collapsing.

IT link: Transformers need **AC only** — a steady DC flux never induces anything, which is why switching-mode PSUs rectify to DC and then chop it back to a changing current before the transformer sees it. The same Faraday law reads a card mag-strip and writes a hard disk.

9 Self & Mutual Inductance

Definition: **Self-inductance:** a coil opposes any change of its *own* current — $L = N\Phi_B/I$, emf = **$-L\,dI/dt$**, energy **$U = \frac{1}{2}LI^2$**. **Mutual:** a changing current in coil 1 induces an emf in coil 2 — $M = N_2\Phi_2/I_1$, $\varepsilon_2 = -M\,dI_1/dt$. Unit: **henry (H)** = V·s/A.

- $L \propto N^2$; a soft-iron core raises it enormously
- A current **cannot change instantly** in an inductor
- **Transformer:** Vs/Vp = **Ns/Np**, Is/Ip = **Np/Ns**

Quantity	Symbol	SI unit
Self-inductance	L	henry (H)
Mutual inductance	M	henry (H)
Magnetic flux	Φ_B	weber (Wb)
Flux density	B	tesla (T)

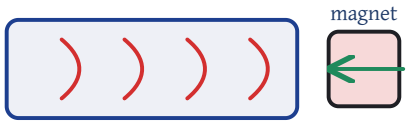


flux through coil 2 caused by coil 1

the transformer: mutual inductance

10 Eddy Currents

Definition: A **bulk conductor** cutting a changing magnetic flux develops closed circulating currents inside the material. Lenz's law makes them **oppose the motion**, so the motion is damped and the energy emerges as **Joule heat** (I^2Rt).



slab of metal

loops circulate inside the whole slab

- Flux must **change**: a solid block heats the moment it moves
- Uses: induction furnace, eddy-current **brake**, metal detector, induction cooker, meter damping
- Core losses are killed by **laminated silicon-steel** laminations

Exam trap: Eddy currents are **not** a single current along a path — every closed loop inside the metal carries one, and all of them oppose the motion: drag, never push.

Worked example ✍️: A 2 H inductor carries 3 A: $U = \tfrac{1}{2}LI^2 = \tfrac{1}{2} \times 2 \times 9 =$ **9 J**. Cutting the current to zero in 0.05 s gives an average opposing emf $L\Delta I/\Delta t = 2 \times 3/0.05 =$ **120 V** — the reason a switch across an inductive load sparks.

11 Maxwell's Equations (1865)

Four equations that **unify electricity, magnetism and light**. Maxwell predicted EM waves in 1865; **Hertz** produced and detected them in **1887–88**.

#	Law	Integral form	Says
1	Gauss (E)	$\oint E\cdot dA = q/\varepsilon_0$	charge is the source of E
2	Gauss (B)	$\oint B\cdot dA = 0$	**no magnetic monopoles**
3	Faraday	$\oint E\cdot dl = -d\Phi_B/dt$	changing B creates a circulating E
4	Ampère–Maxwell	$\oint B\cdot dl = \mu_0 I + \mu_0\varepsilon_0\, d\Phi_E/dt$	current *and* changing E create B

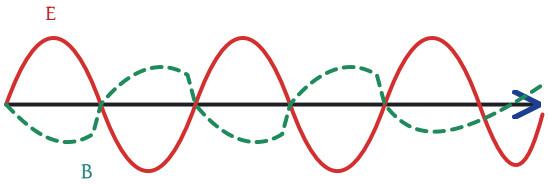
Differential: $\nabla\cdot E = \rho/\varepsilon_0 \cdot \nabla\cdot B = 0 \cdot \nabla\times E = -\partial B/\partial t \cdot \nabla\times B = \mu_0 J + \mu_0\varepsilon_0\, \partial E/\partial t$

Displacement current: $I_d = \varepsilon_0\, d\Phi_E/dt$ — Maxwell's own addition. It lets "current" count as flowing across the **gap of a charging capacitor**, and it makes EM waves possible.

Memory trick: Order **G–G–F–A**: two **divergences** (sources — charge: yes · monopole: no) and two **curls** (circulation — Faraday, Ampère).

12 Electromagnetic Waves

- Speed in vacuum $c = 1/\sqrt{(\mu_0\varepsilon_0)} = 3 \times 10^8$ m/s, independent of frame or source
- **E ⊥ B ⊥ direction of travel**; |E| = cB, so E/B = c
- **Transverse**, self-sustaining, needs no medium
- Carries energy ($S = E \times B/\mu_0$) and momentum
- The whole spectrum shares c: $c = f\lambda$



E and B in step, 90° apart, both ⊥ to the ray travelling to the right

Worked example ✍️: At 100 MHz, $\lambda = 3 \times 10^8 / 10^8 =$ **3 m** (FM band). A 60 Hz mains line has $\lambda = 5 \times 10^6$ m — five thousand kilometres per cycle.

Exam trap: It is the **second** equation ($\nabla \cdot B = 0$) that rules out monopoles. $\nabla \cdot E = \rho/\varepsilon_0$ uses the **volume** density ρ; the integral form uses the **total** charge q inside.

IT link: Free-space wave impedance $Z_0 = \sqrt{(\mu_0/\varepsilon_0)} \approx$ **377 Ω** ($\approx 120\pi\ \Omega$), and a half-wave dipole radiates about **73 Ω** — exactly why Wi-Fi antennas run on 75 Ω coax.

Night-Before-the-Exam Sheet

Constants & formulas:

$e = 1.602 \times 10^{-19}$ C	1 eV = 1.602×10^{-19} J
$k = 1/4\pi\varepsilon_0 = 9 \times 10^9$	$\varepsilon_0 = 8.854 \times 10^{-12}$
$\mu_0 = 4\pi \times 10^{-7}$	$c = 1/\sqrt{(\mu_0\varepsilon_0)} = 3 \times 10^8$
$F = kq_1q_2/r^2$	$E = kq/r^2$
$\oint E \cdot dA = q/\varepsilon_0$	cube face = $q/6\varepsilon_0$
$V = kq/r$	$E = -dV/dr$
$C = Q/V = \varepsilon_0\varepsilon_r A/d$	$U = \frac{1}{2}CV^2 = \frac{1}{2}Q^2/C$
series $C_1C_2/(C_1+C_2)$ · parallel C_1+C_2	$V = IR$ · $R = \rho L/A$
$H = I^2Rt = V^2t/R$	$B = \mu_0 I/2\pi r$ · $\mu_0 I/2R$ · $\mu_0 nI$
$F = qvB \sin\theta$	$F = BIL \sin\theta$
$\Phi_B = BA \cos\theta$	$\varepsilon = -N\, d\Phi_B/dt$
$\varepsilon = BLv$ · $\varepsilon = NBA\omega \sin\omega t$	$U = \frac{1}{2}LI^2$ · Vs/Vp = Ns/Np

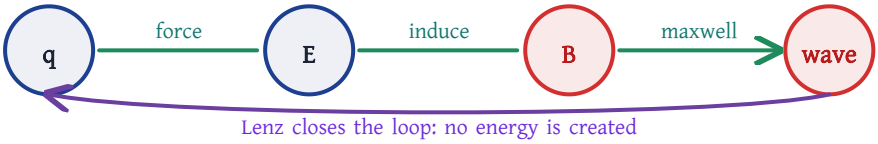
Top 14 one-liners for MCQs

1. Charge is quantised: $q = \pm ne$
8. Cube-face flux = $q/6\varepsilon_0$

2. Charge is conserved in an isolated system
3. $k = 9 \times 10^9$ is $1/4\pi\varepsilon_0$, not ε_0
4. Doubling r divides F by 4
5. $E \propto 1/r^2$ but $V \propto 1/r$
6. 1 T = 10^4 gauss · 1 Wb = V·s
7. E = 0 inside a conductor
9. Infinite sheet: $E = \sigma/2\varepsilon_0$
10. Series capacitance is below the smallest
11. Half L and double A ⇒ R unchanged
12. Constantan / nichrome: $\alpha \approx 0$
13. F = 0 for v ∥ B or a resting charge
14. No monopoles ⇒ $\nabla \cdot B = 0$

Top traps: An emf is volts, not newtons · Lenz opposes the **change**, not the field · $\varepsilon = -N\, d\Phi/dt$ needs the turns count · flux uses the angle to the **normal** · magnetic force does no work · farad/weber/henry do not swap · a transformer needs AC only.

All tricks together: Coulomb is central and inverse-square · sphere→$1/r^2$, line→$1/r$, sheet→constant · E is the downhill slope of V · series capacitance is the tightest link · ρ is the material fingerprint · Fleming's **left** hand for the motor · Faraday: more turns + faster change = more emf · Maxwell order G-G-F-A, two divergences and two curls.



q → E → B → EM wave: the story of the chapter in one line

✓ Practice MCQs (exam pattern, 10 marks)

1. SI unit of electric charge:
   (a) Ampere (b) Volt
   (c) Farad (d) Coulomb
2. Distance doubled, force F →:
   (a) F/4 (b) F/2
   (c) 2F (d) 4F
3. Denies magnetic monopoles:
   (a) $\nabla \cdot E = \rho/\varepsilon_0$ (b) $\nabla \cdot B = 0$
   (c) $\nabla \times E = -\partial B/\partial t$ (d) $\nabla \times B = \mu_0 J + \mu_0\varepsilon_0\, \partial E/\partial t$
4. 100 turns, $\Delta\Phi = 0.04$ Wb in 0.2 s → ε:
   (a) 2 V (b) 0.2 V
   (c) 20 V (d) 200 V
5. +2 μC, +3 μC, 0.3 m apart:
   (a) 0.6 N, repulsive (b) 6 N, repulsive
   (c) 0.06 N, attractive (d) 0.6 N, attractive
6. SI unit of flux density B:
   (a) Weber (b) Henry
   (c) Gauss (d) Tesla
7. E inside a conductor at equilibrium:
   (a) kq/r^2 (b) zero
   (c) σ/ε_0 (d) infinite
8. 3 μF and 6 μF in parallel:
   (a) 2 μF (b) 18 μF
   (c) 9 μF (d) 4.5 μF
9. L doubled, A halved → R is:
   (a) R (b) 4R
   (c) 2R (d) R/2
10. EM wave speed in vacuum:
    (a) 3×10^8 m/s (b) 340 m/s
    (c) 1.5×10^8 m/s (d) 3×10^6 m/s

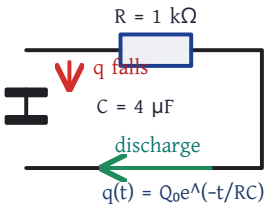
Answer key → 1-d · 2-a · 3-b · 4-c · 5-a · 6-d · 7-b · 8-c · 9-b · 10-a

Written question: A 4 μF capacitor charged to 200 V is connected across a 1 kΩ resistor. Find the charge and energy stored, the time constant, the charge after one time constant, and the heat dissipated.

10 marks

Model answer ✍️:

1. **Stored:** $Q_0 = CV = \mathbf{8 \times 10^{-4}\ C}$; $U = \frac{1}{2}CV^2 = \mathbf{0.08\ J}$ — 3
2. **τ:** $\tau = RC = \mathbf{4\ ms}$ — 2
3. **After one τ:** $q = Q_0 e^{-t/\tau} \rightarrow q(\tau) = Q_0/e = \mathbf{2.94 \times 10^{-4}\ C}$ — 3
4. **Heat:** all of it becomes Joule heat, $H = \mathbf{0.08\ J} = \frac{1}{2}Q_0V$ — 2



charge decays, all energy ends as heat

Before the exam ✓

- ✓ F, E, V: which is 1/r, which $1/r^2$?
- ✓ All four Maxwell equations, both forms
- ✓ Gauss's law: sphere, line, sheet, cube
- ✓ emf from N, ΔΦ, Δt and the sign

Also practise (written):

① Gauss's law to derive $E = kq/r^2$

② emf of a 500-turn coil at 300 rpm in B = 0.2 T